

TRAILWAY – ONLINE TRAIN TICKET APPLICATION

A Project Report

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BONAFIDE CERTIFICATE

Certified that this project report **“TRAILWAY – ONLINE TRAIN TICKET BOOKING APPLICATION”** submitted as part of the CBI9741 – Usability Design of Software Applications course is the bonafide work of **SWETHAA SHANMUGAM R (191401038) & VARSHA S (191401042)** who carried out the project under my supervision.

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ABSTRACT

The Indian Railways (IR) carries over 5.5 lakh people every day in reserved seats. The computerised passenger reservation system (PRS) lets ticket buyers and cancelers buy and sell tickets from any of the 4000 terminals (i.e., PRS booking window all over the countries). With a maximum travel time of 72 hours and a maximum distance of several thousand kilometres, these tickets can be bought or cancelled for excursions beginning in any area in India and ending in any location. In the current project, we will develop a website that will let users obtain information about trains, book and cancel tickets, and determine the exact cost of their tickets to the desired place. People may easily book their tickets online from the convenience of their own homes by clicking a mouse. Using their credit cards, customers may easily and swiftly purchase tickets.

A tool used by the travel and tourism sector to assist online reservations is known as an online booking engine. It makes it easier for customers to make online reservations for services such as hotels, vacation packages, insurance, and flights. Given that aviation is one of the industries with the fastest developing channels, this application is quite important. More than a million people live in India, and our Indian Railways has become a leader in offering high-quality transport services to them. Offering consumers cutting-edge services like online checking, booking, and reservation options has not been a concern for them. With the touch of a mouse at home, the journey may be started using the Indian Railways' online railway booking portal's convenient travel options.

CHAPTER 1

INTRODUCTION

1.1. GENERAL:

Rail travel is one of the most important forms of transportation and has made a substantial contribution to China's transportation industry. Due to China's fast economic development, both the number of railway lines and passengers has been increasing regularly. The problem has been noticed by many people due to the size of the population purchasing train tickets. Electronic commerce may be able to overcome the problem of railroad tickets. The introduction of a new online ticketing system symbolises not only technological innovation but will also, to some extent, enhance train services and handle the complex problem of rail tickets.

The status of their reservations, the availability of tickets for a certain train or destination, details regarding the arrival or departure of particular trains, and other information are frequently required by rail passengers. These questions cannot be answered by customer service departments at train stations during busy hours. The number of reservation desks that are accessible to customers and travellers is quite small. There are generally lengthy wait times on the reservation systems, making it difficult for anybody to book a ticket. The passenger's inquiries cannot now be answered by a call centre.

1.2. MOTIVATION

User Interface Design is essential since it has the ability to increase or decrease the clientele. As a result, problems are reduced, user engagement is increased, functionality is perfected, and a strong relationship between users and the software is created. The previous train ticket booking app's functionality lacked user friendliness, therefore altering the app's look would improve users' experiences. Users will be able to look for train availability, precise fares, arrival and departure times, and purchase tickets using debit, credit, or master cards. If users need to cancel their tickets after booking them, they may do so with ease as well.

1.3. OBJECTIVE

This Railway application project's goal is to create software that completely automates the railway ticketing procedure -

- To compile a train database
- To look up train arrival and departure times, as well as the distance between the sources.
- And the destination.
- To confirm the ticket's availability.
- To determine the fare.
- To purchase a ticket.
- If necessary, to cancel the ticket.

1.4. EXISTING DESIGN

The existing system for online train Ticketing applications helps users to understand the working of Trailway app. These applications include a variety of

- The interface is too cluttered, making it difficult for the user to concentrate on a single
- Task.
- Language barriers can be problematic. MMT only provides its services in Hindi and English.
- Accessibility is limited because none of the competitors that we looked at above support
- Voice search.
- There are no manuals or resources on the various jargons used in these applications to specify classes.
- In case of an emergency, users are unable to use SOS services.

1.5. PROPOSED DESIGN

The working of this application is made like the user with a great deal of freedom, making walking on it simple These Includes

- Good UI
- Not cluttered and user friendly
- SOS available
- Voice options.

The user will be able to see the offers and return booking details on the home page which makes it easier for users to navigate easily and all the drawbacks are rectified.

1.6 TOOLS AND TECHNOLOGIES

1.6.1 FIGMA

Figma is a cloud-based design tool that is similar to Sketch in functionality and features, but with big differences that make Figma better for team collaboration. For those sceptical of such claims, we'll explain how Figma simplifies the design process and is more effective than other programs at helping designers and teams work together efficiently. Figma works on any operating system that runs a web browser. Macs, Windows PCs, Linux machines, and even Chromebooks can be used with Figma. It is the only design tool of its type that does this, and in shops that use hardware running different operating systems, everyone can still share, open, and edit Figma files.

Since Figma is browser-based, teams can collaborate as they would in Google Docs. People viewing and editing a file are shown at the top of the app as circular avatars. Each person also has a named cursor, so tracking who is doing what is easy. Clicking on someone else's avatar zooms to what they are viewing at that time.

Figma has gone further by providing transitions between frames. Figma's simple prototyping feature eliminates the need for another tool that does slideshow style prototyping.

CHAPTER 2

PROJECT DESCRIPTION

The user of this programme can log in using either their Google account or their phone number and an OTP will be received. After doing so, they can search the trains that are currently running based on factors like class, origin, and destination. The available trains are presented with all these filters applied, and the After that, the user can reserve a train based on their preferences. The user can also view the status of their ticket, including whether it is 100% confirmed or on the waiting list, and its dependability. The users can then, if necessary, add their preferences. All the relevant information, including the fee and train information, are reviewed. You can then pay for your ticket using any of the electronic payment methods, including UPI, net banking, and so on. The ticket is also available for download.

For the users to use this application more conveniently, there are several settings available. It was produced using the UI/UX tool Figma. For instance, even if we don't receive an OTP during the sign-up procedure, we don't need to proceed to the first page; instead, the screen just refreshes and a new OTP is generated, making it convenient and reducing needless activities. As a result, time is saved.

This application has a voice option so that users can access it as well, allowing it to be used by a wider spectrum of users regardless of their level of understanding. The user can search all of the neighbouring stations using the search bar, which is also included.

CHAPTER 3

IMPLEMENTATIONS AND RESULTS

3.1. PRODUCT APPRECIATION ASSIGNMENT

3.1.1. Ease of Use

The UI layout of this application primarily focuses on the easy accessibility to the users and thereby facilitating the same. Trailway is a ticket booking application and makes the user very convenient to use the app without any obstacles. Thus, following are some criteria that were looked into while designing this application.

- a. Only the necessary information is available on the screen and is not cluttered.
- b. The icons and other navigation bars can easily be accessed.
- c. The layout is properly designed with interactive elements.
- d. Users can interact with the application via various options available.

In this application, there is a voice option available so that the user can also access the application in this manner making it to reach a wider range of users irrespective of their knowledge. Additionally, there is also a search bar so that the user can search all the nearby stations available.

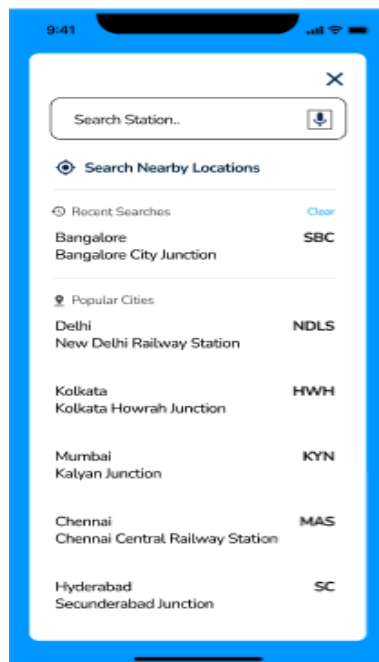


Fig 1- Image describing Ease of use in applications

3.1.2. Functionality

In this application, the user can either sign in via their google account or their phone number with the OTP and then can search the trains available according to their feasibility such as the type of class, from location and the destination. With all these filters, the available trains are listed and the user can then book the train according to the preference. Also, the user can see their ticket and its reliability like whether it is 100% confirmed or is in the waiting list. Then

the users can add their preferences if needed. A review is presented of all the necessary details with the fare and the train details. Then you can book your ticket by paying in any of the electronic modes such as UPI, net banking and so on. And you can also download the ticket.

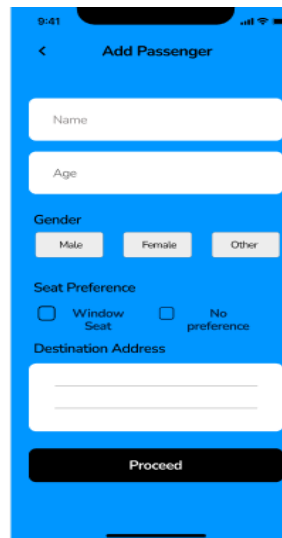
A screenshot of a mobile application interface titled "Add Passenger". The form is set against a blue background. It includes input fields for "Name" and "Age". Below these is a "Gender" section with three buttons: "Male", "Female", and "Other". The "Seat Preference" section has two checkboxes: "Window Seat" and "No preference". At the bottom is a "Destination Address" input field and a black "Proceed" button.

Fig 2 - Image describing Functionality in applications

3.1.3. Ergonomics

There are a lot of options available in this application in order for the users to use it in an easier manner. It is created using Figma which is an UI/UX tool. For instance, even if we do not receive OTP during the sign-up process there is no necessity to go through the first page, it just reverts back to the screen and a new OTP is generated making it convenient and unwanted activities are reduced. Hence, this saves time.

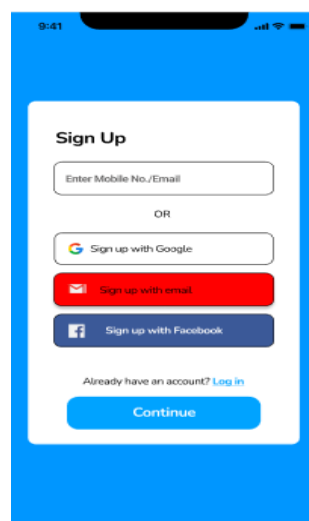
A screenshot of a mobile application interface titled "Sign Up". The screen has a blue background with a white central card. The card contains an input field for "Enter Mobile No./Email", followed by the text "OR". Below this are three social login buttons: "Sign up with Google" (with a Google logo), "Sign up with email" (with a red envelope icon), and "Sign up with Facebook" (with a Facebook logo). At the bottom of the card, there is a link "Already have an account? Log in" and a blue "Continue" button.

Fig 3 - Image describing Ergonomics in applications

3.1.4. Aesthetics

Aesthetics is one of the important aspects in a design process. The outline of any application makes the users want to use it.

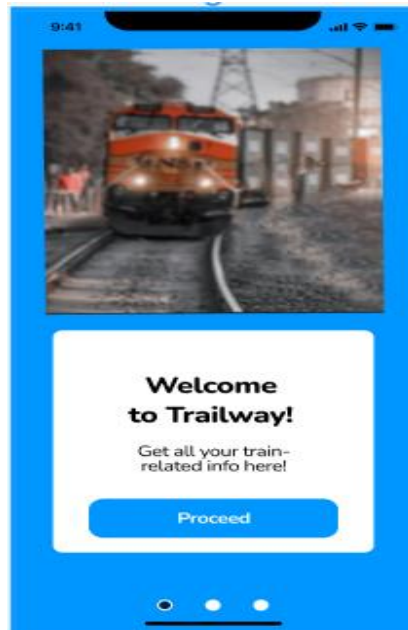


Fig 4 - Image describing Aesthetics in applications

3.2. HEURISTIC EVALUATION

3.2.1 Visibility of system status

“The system should always keep users informed about what is going on, through appropriate feedback within a reasonable amount of time.” In this application, there are several options available in each web page indicating several actions that the user can communicate with the system making it more interactive.

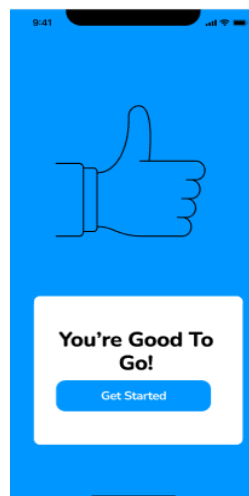


Fig 5 - Image describing System status in applications

3.2.2 Match between System and the Real World

“The system should speak the users' language, with words, phrases and concepts familiar to the user, rather than system- oriented terms. Follow real-world conventions, making information appear in a natural and logical order.”

Here, there are a lot of icons that match the real-world objects adding some additional context. In the page given below, there are three icons:

- Location icon
- Calendar icon
- Help icon

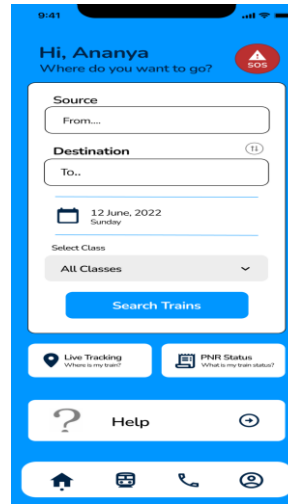


Fig 6 - Image describing between system and real world in applications

3.2.3 User Control and Freedom

Users often perform actions by mistake. They need a clearly marked "emergency exit" to leave the unwanted action without having to go through an extended process. Here, there is a clear option wherein the user can search for locations and can easily search for other locations by just clicking the clear option and searching for new options. There is no necessity to go back to the home page and then follow the same set of options to land this option. This saves the time of the users.

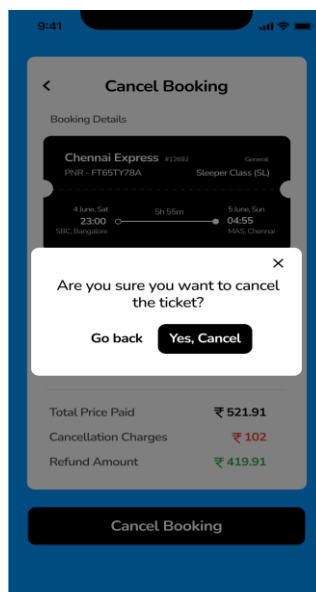


Fig 7 - Image describing User control and freedom in applications

3.2.4. Consistency and Standards

Users should not have to wonder whether different words, situations, or actions mean the same thing. Follow platform and industry conventions. The button design is consistent across the application, creating an understandable design pattern and the user can just click on the button which will process some activity and move on to the next process.



Fig 8 - Image describing Consistency and standards in applications

3.2.5 Error Prevention

“Even better than good error messages is a careful design which prevents a problem from occurring in the first place. Either eliminate error-prone conditions or check for them and present users with a confirmation option before they commit to the action.”



Fig 9 - Image describing Error prevention in applications

3.2.6 Recognition Rather than Recall

“Minimize the user's memory load by making objects, actions, and options visible. The user should not have to remember information from one part of the dialogue to another. Instructions for use of the system should be visible or easily retrievable whenever appropriate.”



Fig 10 - Image describing Recognition rather than recall in applications

3.2.7 Flexibility and Efficiency of Use

“Accelerators — unseen by the novice user — may often speed up the interaction for the expert user such that the system can cater to both inexperienced and experienced users. Allow users to tailor frequent actions.”

3.2.8. Minimalistic

Our application consists of a separate page for Live tracking, contact us and profile which provides user accessibility when they need to search the application every time. The user can see live tracking and also can contact us if there is any problem or check their details in their profile.



Fig 11 - Image describing Minimalistic in applications

3.2.9. Help users recognize, diagnose and recover from errors:

Error messages should be expressed in plain language (no codes), precisely indicate the problem, and constructively suggest a solution.

3.2.10. Help and documentation

Even though it is better if the system can be used without documentation, it may be necessary to provide help and documentation. Any such information should be easy to search, focused on the user's task, list concrete steps to be carried out, and not be too large.

3.3 USER-CENTERED DESIGN LIFECYCLE

4D is a user-centric and iterative process we could use in digital design and development. The phases are:

Discovery	— understanding the problem(s) and developing insights
Design	— the area to focus upon
Develop	— potential solutions to the problem(s)
Deliver/ Implement	— solutions that work

3.3.1 Discovery - Understanding the problem statement

A discovery is a preliminary phase in the UX-design process that involves researching the problem space, framing the problem(s) to be solved, and gathering enough evidence and initial direction on what to do next. Discoveries do not involve testing hypotheses or solutions.

There are many different types of activities that could be carried out in a discovery.

- Exploratory Research
- Stakeholder Interviews
- Workshops

Booking train applications is the foremost in every situation no matter how the environment is. Railway helps to book the tickets in an easier manner having a very simple UI and is user friendly. It is not very cluttered and the icons are very well displayed to be understood by anyone who wishes to use the app. The users can also book the type of class they wish to opt for and other necessary information.

We conducted brainstorming sessions in the classrooms among the students to identify the problems faced by them while booking the tickets and their reactions to it. The below are the points identified while researching the problem:

- The booking process is usually very time consuming. This is the most popular opinion among the students because they feel that it takes a longer time to search for trains and they opt for better modes of transportation which require less time like buses and cabs for shorter distances.
- A lot of train applications will take a lot of time to receive the refund and sometimes you don't get the full cash back if you cancel it very near to the date of booking. Sometimes the user might tend to forget the date of booking and he/she wants to cancel it but because of the fear of cancellation they might not want to use the application again.
- Many applications do not have a window seat reservation and some individuals prefer to travel with their mode of preference and if they are not satisfied with their seat arrangement, they tend to cancel it and don't use the application regularly.

PERSONAS:

A user persona is a representation of the goals and behaviour of a hypothesized group of users. In most cases, personas are synthesized from data collected from interviews with users. User personas are archetypical users whose goals and characteristics represent the needs of a larger group of users. Usually, a persona is presented in a one or two-page document (like the one you can see in the example below). Such 1–2-page descriptions include behaviour patterns, goals, skills, attitudes, and background information, as well as the environment in which a

persona operates. Designers usually create user persona templates, which include a few fictional personal details to make the persona a realistic character (e.g. quotes of real users), as well as context-specific details (for example, for a banking app it makes sense to include a persona's financial sophistication and major expenses).

A deep understanding of a target audience is fundamental to creating exceptional products. User personas help a product team find the answer to one of their most important questions, "Who are we designing for?" By understanding the expectations, concerns, and motivations of target users, it's possible to design a product that will satisfy users' needs and therefore be successful.

PERSONA - 1

We get that the user wants to be able to effortlessly purchase a train ticket whenever she wants to and wherever she can. She has already used a number of train ticketing applications, but she feels that she needs one that will make it more comfortable for her to do so. As a result, this app will serve as a conduit for both her learning and buying train tickets. The user also feels that using this software has been a positive experience.

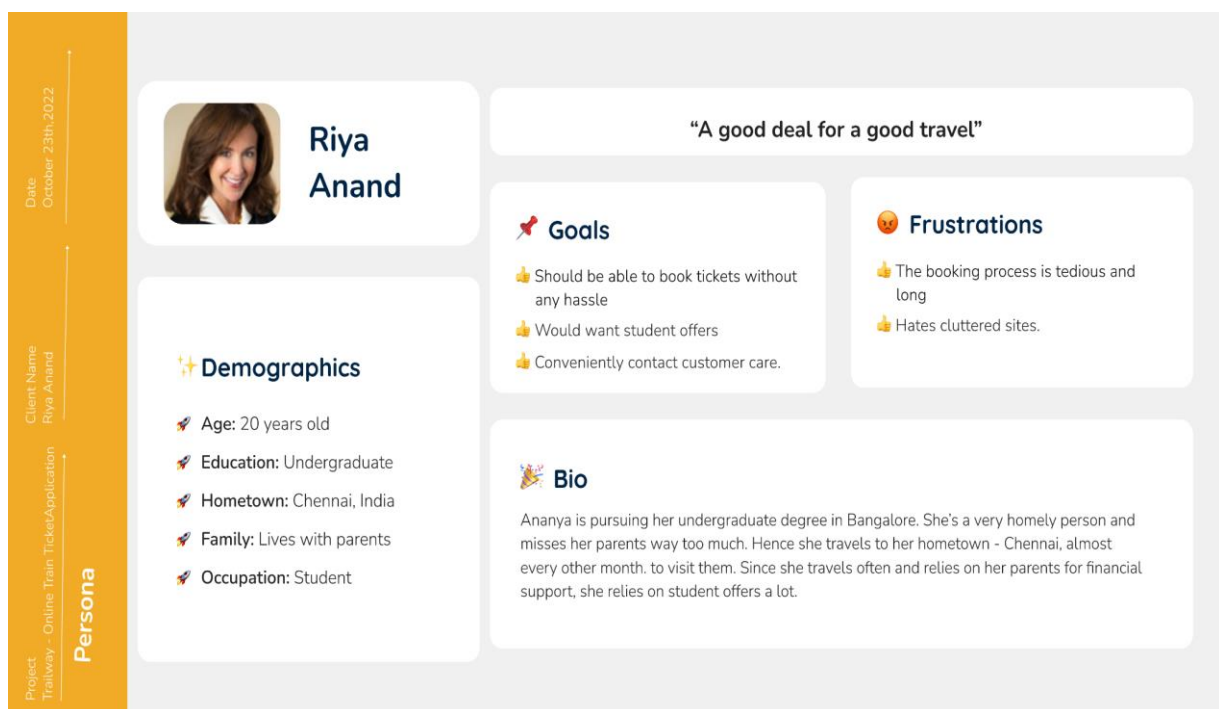


Fig 12 - Image describing Persona-1 in applications

PERSONA - 2

We get that the user wants to be able to effortlessly purchase a train ticket whenever he wants to and wherever he can. He has already used a number of train ticketing applications, but he feels that he needs one that will make it more comfortable for him to do so. As a result, this app will serve as a conduit for both learning and buying train tickets. The user also feels that using this software has been a positive experience.

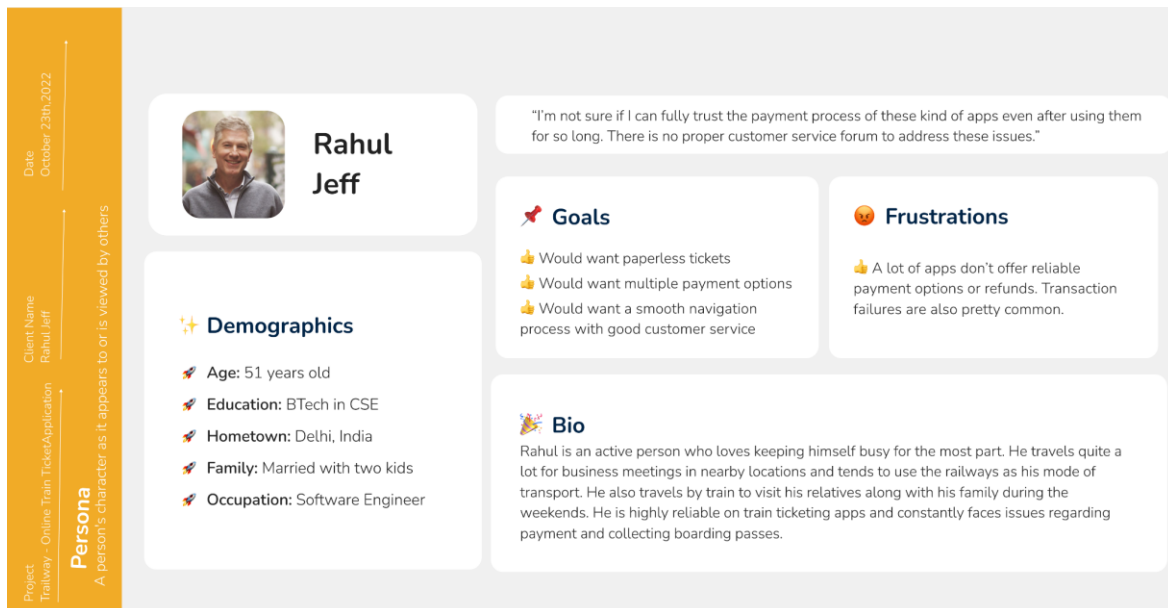


Fig 13 - Image describing Persona-2 in applications

Drafting a User Scenario - Describe the scenario

User A - Ananya

Goals

- Should be able to book tickets without any hassle
- Would want student offers.
- Conveniently contact customer care.

Frustrations

- The booking process is tedious and long.
- Hates cluttered sites.

User B - Guru

Goals

- Would want paperless tickets.
- Would want multiple payment options.
- Would like a smooth navigation process with good customer service.

Frustrations

- A lot of apps don't offer reliable payment options or refunds. Transaction failures are also pretty common.

Riya opens the app and uses it to search for preferred dates as well as to purchase the train tickets that are offered on the booking page. She can easily get a train ticket to Chennai without needing the financial assistance of her parents. This app has numerous student deals, which is quite helpful.

Rahul opens the app and uses it to search for preferred dates as well as purchase the train tickets that are offered on the booking page. He can easily get a train ticket for his business meetings

in a nearby location, and he also likes to visit his relatives along with his family on weekends, and this app has reliable payment and refunds.

3.3.2 DEFINE:

The problems are thereby identified in the discovery stage and the problem statement is identified in the define stage from the insights developed in the discovery phase. After empathizing with the intended users, problem statements can be made which is to be solved in the forthcoming stages.

To be more specific, the define stage is dedicated to defining the problem: what user problem will you be trying to solve? In other words, what is your design challenge?

The defined stage ensures you fully understand the goal of your design project. It helps you to articulate your design problem, and provides a clear-cut objective to work towards. Without a well-defined problem statement, it's hard to know what you're aiming for.

A problem statement, or point of view (POV) statement, frames this problem (or need) in a way that is actionable for designers. It provides a clear description of the issue that the designer seeks to address, keeping the focus on the user at all times. A good problem statement is human-centered and user-focused. Based on the insights you gathered in the empathize phase, it focuses on the users and their needs—not on product specifications or business outcomes.

Here, the problem statement must address the following questions:

1. **Who is experiencing the problem?** In other words, who is your target user; who will be the focus of your problem statement?
Users who are traveling by trains regularly and those who feel dissatisfied with the poor interface of the IRCTC application.
2. **What is the problem?** Based on the observations you made during the empathize phase, what are the problems and pain-points that frequently came up? What task is the user trying to accomplish, and what's standing in their way?
 1. A proper interface which is not cluttered.
 2. A voice search option enhances the accessibility.
 3. SOS service to be made available to the users in case of emergency.
3. **Where does the problem present itself?** In what space (physical or digital), situation or context is the user when they face this problem? Are there any other people involved?
 1. Online users who wish to book tickets on any occasions.
 2. Users who do not want to stand in the queue and prefer online mode.
4. **Why does it matter?** Why is it important that this problem be solved? What value would a solution bring to the user, and to the business?
 - Helps user's book tickets efficiently without any problem.
 - User-friendly
 - Can be accessible any time without any restrictions.

TASK FLOW

Task analysis is the process of learning about ordinary users by observing them in action to understand in detail how they perform their tasks and achieve their intended goals. Tasks analysis helps identify the tasks that your website and applications must support and can also help you refine or re-define your site's navigation or search by determining the appropriate content scope. It's important to perform a task analysis early in your process, in particular prior to design work.

TASK FLOW ANALYSIS:

Step 1: Gain an understanding of the user

The user needs an app that matches with the real-world scenario and also provide cutting-edge technology-based courses for gaining additional knowledge apart from the regular academic schedule which is also tracked on this application

Step 2: Drafting a User Scenario - Describe the user

User A - Ananya

- Age: 20 years old
- Hometown: Chennai, India
- Occupation: Student
- Ananya is pursuing her undergraduate degree in Bangalore.
- She travels to her hometown - Chennai, almost every other month to visit them. Since she travels often and relies on her parents for financial support, relies on student offers a lot.

User B - Guru

- Age: 51 years old
- Hometown: Delhi, India
- Occupation: Software Engineer
- Guru is a Software Engineer native in Delhi India.
- He travels quite a lot for business meetings in nearby locations and tends to use the railways as the mode of transport. He also travels to visit his relatives along with his family during weekends.

Step 3: Drafting a User Scenario - Describe the scenario

User A - Ananya

Goals

- Should be able to book tickets without any hassle
- Would want student offers.
- Conveniently contact customer care.

Frustrations

- The booking process is tedious and long.
- Hates cluttered sites

User B - Guru

Goals

- Would want paperless tickets.
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- Would like a smooth navigation process with good customer service.

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Riya opens the app and uses it to search for preferred dates as well as to purchase the train tickets that are offered on the booking page. She can easily get a train ticket to Chennai without needing the financial assistance of her parents. This app has numerous student deals, which is quite helpful.

Rahul opens the app and uses it to search for preferred dates as well as purchase the train tickets that are offered on the booking page. He can easily get a train ticket for his business meetings in a nearby location, and he also likes to visit his relatives along with his family on weekends, and this app has reliable payment and refunds.

Step 4: Sketching a Task Flow

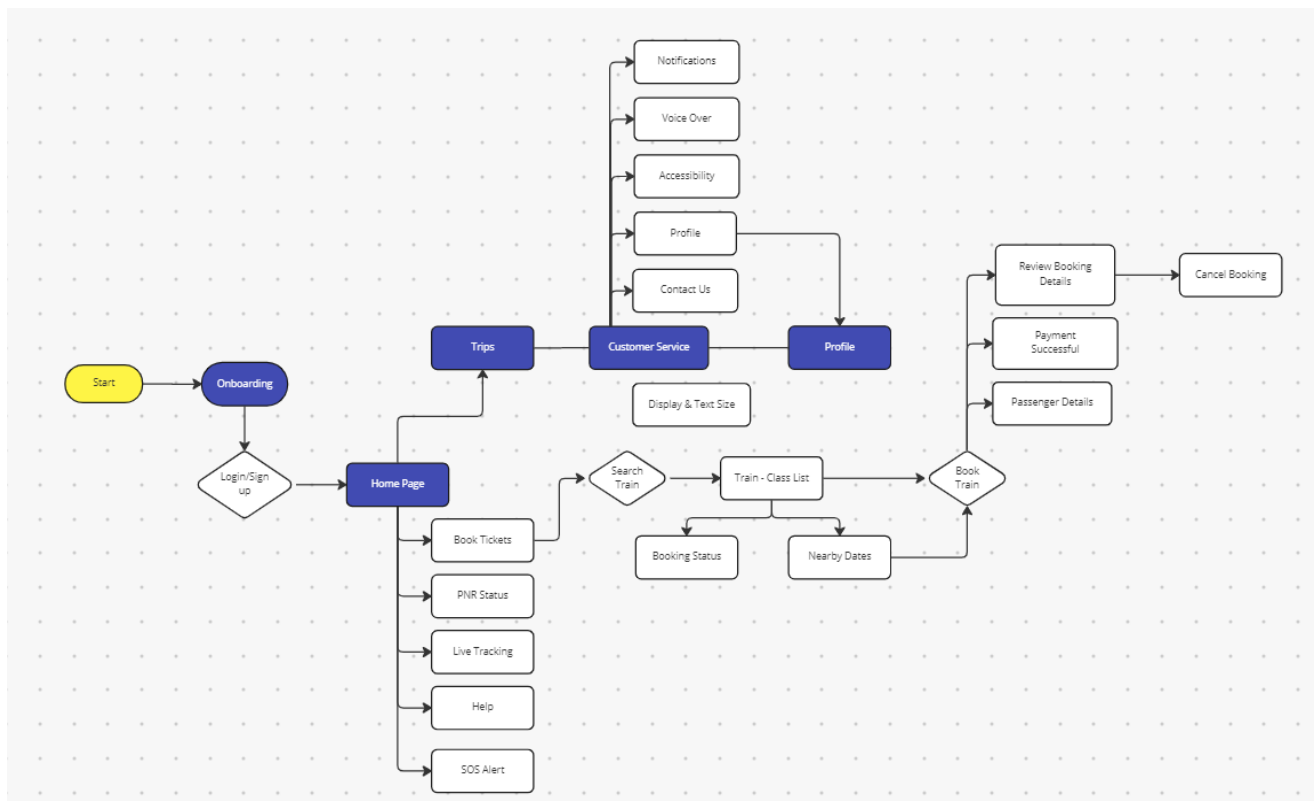


Fig 14 - Image describing Task flow in applications

3.3.3 DESIGN:

The design phase is carried out for the train app by listing out the screens and activities performed and the requirements in a flow to understand the process of making the UI design for the application.

Here, there is a proceed button in the splash screen wherein the interested users can click in order to use the application. Following there is a pop-up screen showing whether to receive the notifications and to turn on the location are not. Then the users can sign up using their mobile number or their email id and then receive the otp and can have their access to this application. Then, the trains can be searched using the listed filters like the source and destination locations, type of class etc. You can also see the detailed information of the ticket and can see the reservation facilities and the seat allocation also. SOS option is also available making it more feasible to the users. The whole flow is aimed to be done with a minimalistic and effective UI design approach.

WIRE FRAMING AND PROTOTYPE

Wire framing

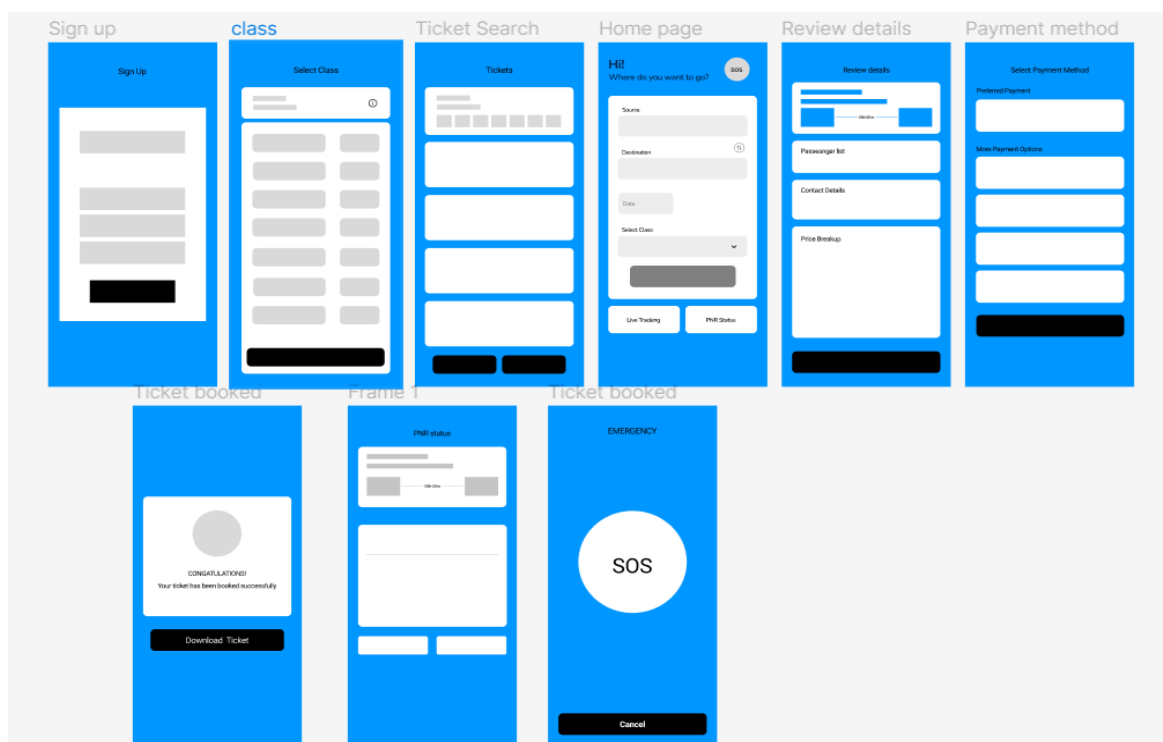


Fig 15 - Image describing Wire framing of applications

Prototyping

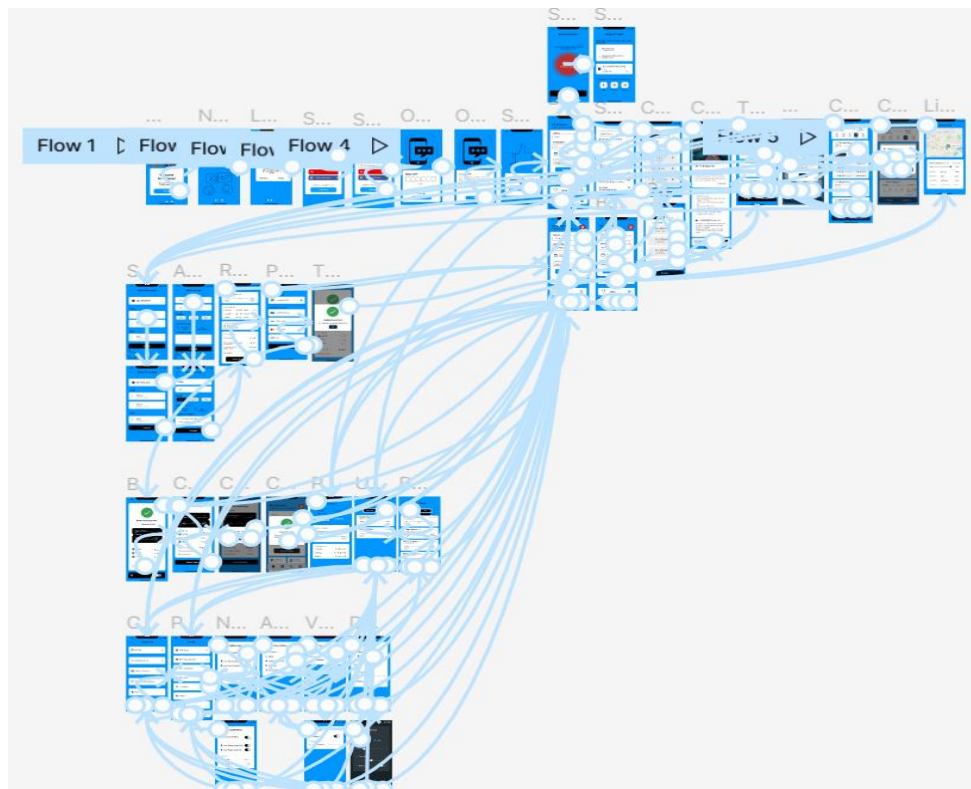


Fig 16 - Image describing Prototyping of applications

PROTOTYPING

Prototyping is an experimental process where design teams implement ideas into tangible forms from paper to digital. Teams build prototypes of varying degrees of fidelity to capture design concepts and test on users. With prototypes, you can refine and validate your designs so your brand can release the right products.

Prototyping is of two types:

- 1) Paper prototypes
- 2) Digital prototypes

1. Paper prototypes:

Paper prototyping is the process of developing ideas and designing user flows using hand-sketched “screens” that represent a digital product. Paper prototypes test on a high-level user experience rather than interaction design.

Paper prototypes are low-fidelity because they don’t have any functionality. For this reason, paper prototypes designers rarely share paper prototypes outside of the department. The primary goal of paper prototyping is to map our information architecture and visualize user flows.

Advantages:

- Fast — You can sketch a prototype in minutes, which is why paper works so well for testing lots of ideas. You can draw a prototype quickly (even during a brainstorming meeting), so you haven't wasted more than a few minutes if an idea falls flat.
- Inexpensive — You only need a marker pen and paper to create prototypes, making the process cheap and accessible.
- Team-building — Paper prototyping is a collaborative effort, and often teams have fun coming up with fresh ideas. It's a fantastic team-building exercise, and these free-thinking sessions often inspire creativity.
- Documentation — Team members can keep physical copies of paper prototypes, notes, and to-dos for quick reference during future iterations.

Disadvantages

- Unrealistic — No matter how skilled the art or craftsmanship, paper prototypes will never be more than hand-drawn representations of a digital product. So, while they're quick to draw, paper prototypes produce little or no results when doing user testing.
- False positives — sometimes, paper prototypes don't validate ideas properly. What seems like a good idea on paper might not work effectively in a digital wireframe.
- No gut reactions — Paper prototypes rely on the user's imagination, adding a break between seeing the stimulus and responding to it. That "gut" reaction is crucial for a successful UX.

2. Digital Prototyping:

Digital prototyping is an exciting part of the design process. Prototypes start to resemble the final product allowing teams to test and validate ideas.

There are two types of digital prototypes:

- Low-fidelity prototypes: a user flow using wireframes
- High-fidelity prototypes: a user flow using mockups

Low Fidelity Prototypes

Low-fidelity prototypes, for example, are simple and low-tech concepts. All you need to get started is a pen and paper. The goal is to turn your ideas into testable artifacts that you can then use to collect and analyse feedback in the early stages. Low-fidelity prototypes allow research teams to outline basic user flows and information architecture.

Designers build prototypes using design tools like Figma, Adobe XD, and others. Sometimes non-designers, from product teams use Power point or Google Slides to simulate user flows.

Low Fidelity Prototype Frames



Fig 17 - Image describing Low fidelity prototype of in applications

User when using the low fidelity prototype, they face the following problems:

- The users could not understand the flow of the booking process.
- The user cannot confirm the authentication process.
- The model does not have a good user interface and it is too cluttered with too much text making it difficult.
- The model's user interface is poor because there are no colors nor graphics to provide an effective user interface.
- The model doesn't have multiple payment options to book the tickets.
- The model doesn't give any transaction details that the user has paid.
- There are no icons or buttons used.
- The model doesn't have any match between the real world and icons used.
- There are no details about the waiting list confirmation.
- The search and the filters button available in the home page is not effective.
- The home page is inefficient and has many blank spaces which confuses the users.

- The interface is not minimalistic and has big difficult words which makes understanding the model to the user difficult and not wanting the user to use it again.
- The model lacks adaptability and effectiveness.

These are the pain points that are gathered from the feedback that users gave after using the model. These pain points are then analysed by the team and the model is redesigned based on the feedback obtained from the user.

By redesigning the model, the above-mentioned problems are addressed and the model is developed with user friendliness and in an efficient and effective way.

High-Fidelity Prototype

High-fidelity prototypes are highly functional and interactive. They are very close to the final product, with most of the necessary design assets and components developed and integrated. Hi-fi prototypes are often used in the later stages to test usability and identify issues in the workflow. High-fidelity prototypes go into more detail, testing user interfaces, interactions, and how usability participants interact with a product.

Designers build prototypes using design tools like Figma, Adobe XD, and others. Sometimes non-designers, from product teams use Power point or Google Slides to simulate user flows.

Advantages

1. The higher the fidelity, the better the interaction.

Wireframes do not give an actual output – the user will never see them again. In fact, wireframes are those platforms from which visual designers and developers make a real output.

2. Clients get a better overview of the final design, before going through the entire SDLC.

High-fidelity wireframes give a better idea of how the platform will look before going through the final design stage and the entire software development lifecycle (SDLC). Whereas, it becomes difficult with low-fidelity wireframes for the client to visualize what the end product would look like. High-fidelity wireframes do not require much imagination or visualization from the client – in order to understand the concept.

3. High-fidelity wireframes are more detailed than low-fidelity wireframes.

High-fidelity wireframes go into more details like the use of programs, colors and logos whereas low-fidelity wireframes use placeholders or, say, real objects such as handmade sketches, which may sometimes look shoddy.

The benefits of high-fidelity wireframes include:

- They show a lot more specifics
- They provide a better sense of what real UI elements might look like
- They provide easier client-developer interactions
- The industry will never fully be on board with high-fidelity wireframes only.

Disadvantages

1. They can create client confusion
2. They can curb a client's creativity
3. They can cause clients to think they are a perfect prototype when they are still very much a work in progress.
4. They cost more time and money from a developer and the client may not be satisfied until the wireframe is exactly how they want the actual site to look and function

High-Fidelity Prototype Frames

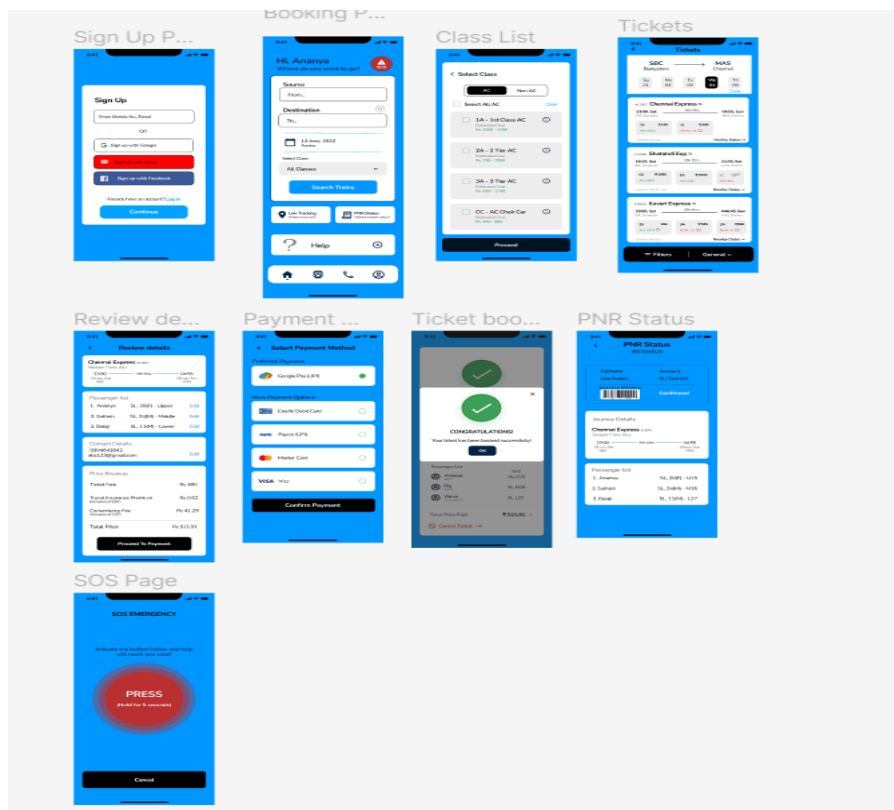


Fig 18 - Image describing High fidelity prototype of in applications

3.3.4 IMPLEMENTATION:

The designs are implemented using Figma and designs structures are as follows : Font Used: Poppins, Color Shades Used: Error, Warning, Success, Secondary, Primary, Neutral Shades: White - 0, Black - 100, Green - 50 , Text Style: Semibold, Light, Regular (Medium / Large / Regular) , Background Color (Hex): 3AD321

The interaction between the pages is easily implemented using the Figma tool. Connection is created by navigating to the Prototype tab of the right sidebar and selecting a layer or object for the connection's hotspot.

Once you have created a connection, you can define the type of interactions users can have with it. You're able to create multiple interactions from the same layer or object.

Figma is a cloud-based design tool that is similar to Sketch in functionality and features, but with big differences that make Figma better for team collaboration. For those skeptical of such claims, we'll explain how Figma simplifies the design process and is more effective than other programs at helping designers and teams work together efficiently.

Figma works on any operating system that runs a web browser. Macs, Windows PCs, Linux machines, and even Chromebooks can be used with Figma. It is the only design tool of its type that does this, and in shops that use hardware running different operating systems, everyone can still share, open, and edit Figma files.

The modules of the model is as follows:

Verification

In order to use Trailway, the first step is to sign up with an email id or your google account. We can also sign in using our mobile number and receive an OTP in order to use the application. This is a very secure step and very pivotal as it is the rudimentary stage involved in this application.

After signing up with correct credentials, the user can use Trail Way to book the tickets and travel to the desired location. If the user does not provide correct information, they cannot login and use it as it may cause a problem in verification.

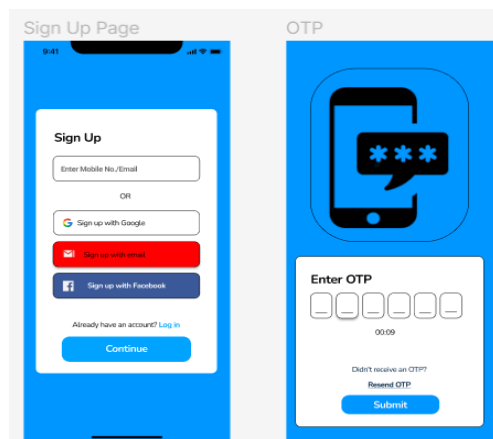


Fig 19 - Image describing Verification page for applications

Home page

There is a separate login assigned to the user when you can book the application accordingly. When the user signs in to the application the user can view the source and destined locations and apply filters and get only the required information according to the filters. Users can also choose the specified date and choose the required accordingly. Also live tracking is there to track the trains' location and can also see the waiting list percentage.

Filters help to scrutinize the process and only get the desired output.

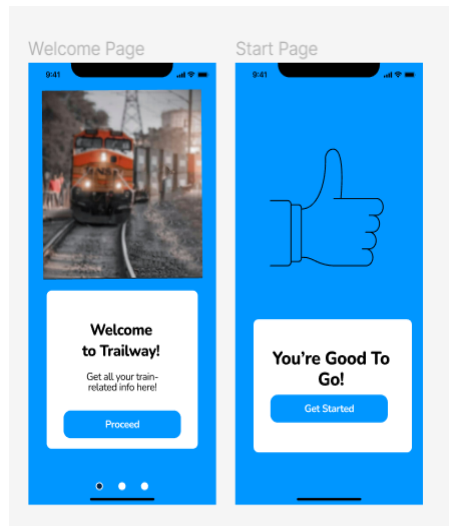


Fig 20 - Image describing home page for applications

Ticket details page

The user gets to know all the ticket details including the train name, date, type of class(sleeper, First class) and Confirmation changes in a box with two buttons namely cancel and proceed where in the user can cancel the application or proceed further. Ticket cancellation is also possible with minimal cancellation charges and the refund amount will be provided to the user within a few days. The user can also prevail additional benefits like choosing the seat type and also provide their personal details while booking the train application.

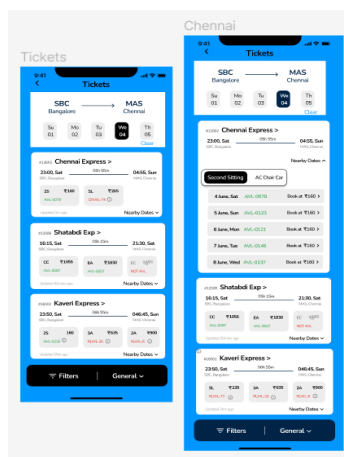


Fig 21 - Image describing Ticket detail page for applications

Menu Functionalities

The train application provides additional benefits such as a Voice over option wherein the user can access the application via voice making it convenient to all types of user, SOS Alert, Live Tracking, Seat Status, Profile and contact details, Sound settings, Accessibility and Display & Text size(Brightness & Bold Size).

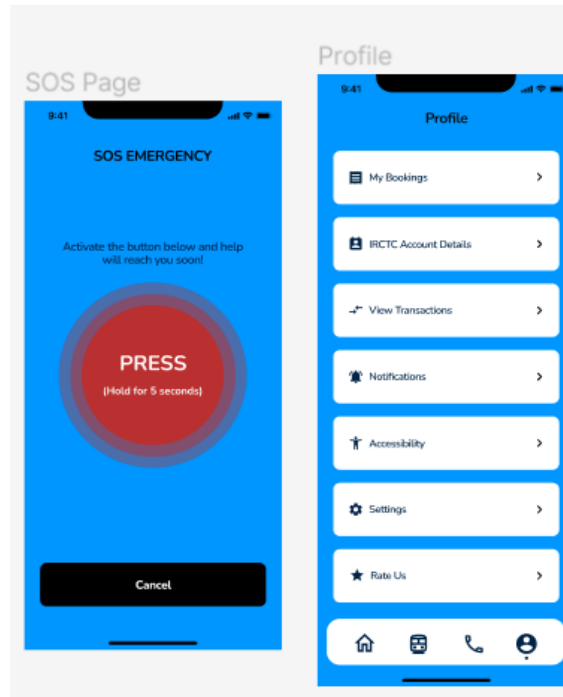


Fig 22 - Image describing Menu page for applications

3.3.5 USABILITY TESTING:

Usability testing, also known as User Experience (UX) testing, is a testing method for measuring how easy and user-friendly a software application is. A small set of target end-users, use software applications to expose usability defects. Usability testing mainly focuses on the user's ease of using the application, flexibility of application to handle controls and ability of application to meet its objectives.

LIKES • Easy to use • User-friendly • Not cluttered • High accessibility	CRITICISM • Internet connectivity • No web app • No access for web application
QUESTIONS • Will the app be in various languages? • Is the SOS option available for emergency situations? • Can we track the train's location simultaneously while waiting?	IDEAS • Voice assistant for better usage • User preference can be added • Paying the ticket on service

3.3.6 RESULTS

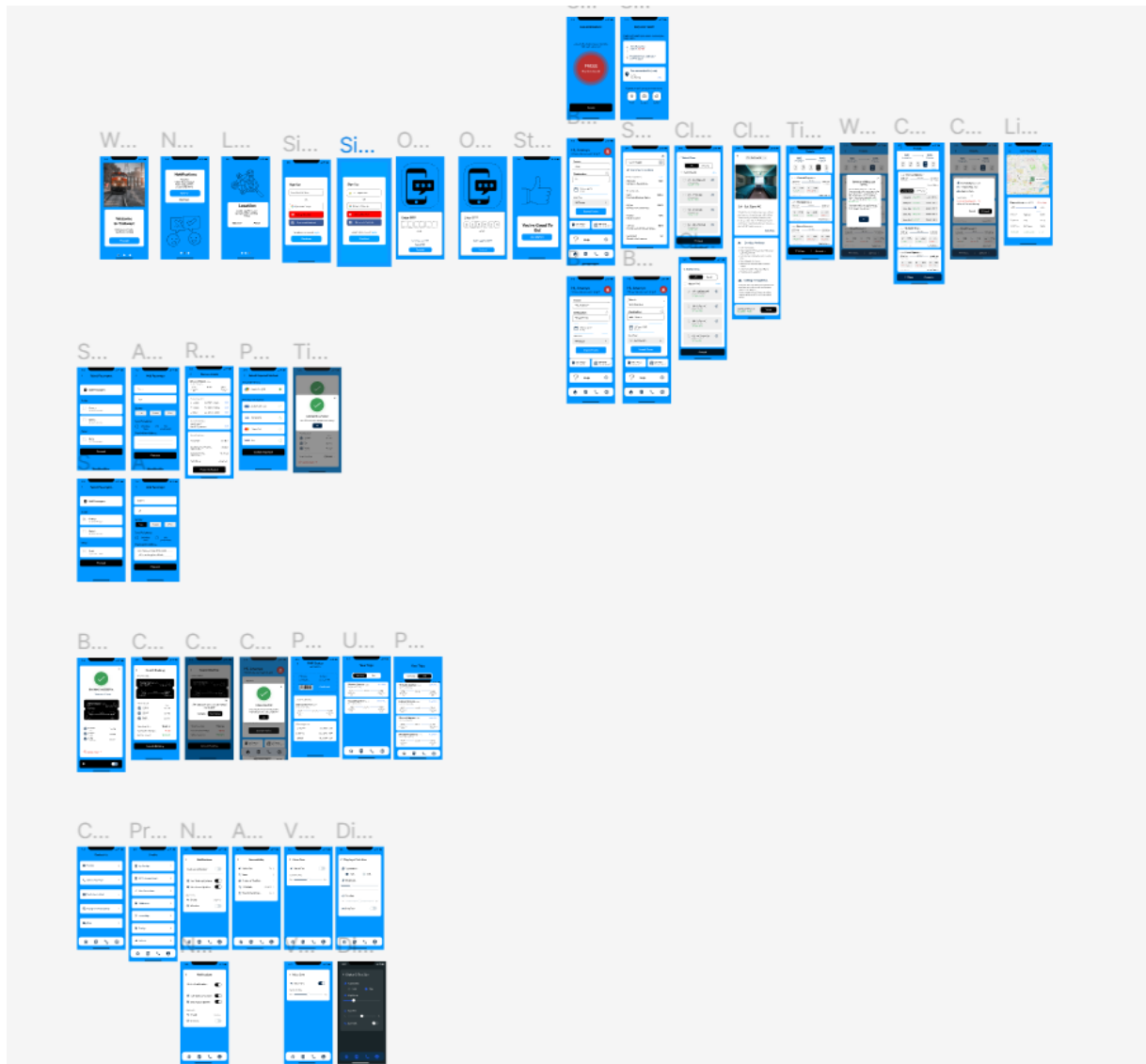


Fig 23 - Image describing Implementation of applications

CHAPTER 4

CONCLUSION

Nowadays, every task must be completed quickly and easily in order to save time and effort while yet producing a useful report.

A customer can book a ticket online, around-the-clock, from anywhere in the world using an online train ticket booking system. Customers can interact with the website for ticket reservations to get any additional information they require. Successful development of an online system for purchasing tickets. Likewise, satisfactory system performance is found.

The application has a good user interface. The cost can be brought down and the efficiency raised through the use of this application. Customers can choose among a variety of procedures. Customers can book tickets, check the status of a train using the application, choose their preferred source station and destination, select seats, and choose the date and time of departure. As you get at the station you can pay through the site. Online ticket buying systems therefore cater to both internal and external audiences.

As a result, they can encode, record, and generate tickets more easily and with less effort.

CHAPTER 5

FUTURE ENHANCEMENT

To make the voice search option in many security languages available in many languages, and to include tracking of the nearby autos after arrival. To include food serving purposes to the users, as well as to introduce interactive icons for easy access to a range of foods and drink options.

CHAPTER 6

LEARNING OUTCOMES

The app lends itself naturally to designs, but because it acknowledges the mythology and cultures that gave rise to the naming of constellations, it may also be used in other contexts. Find websites that are free to use and that we may use with or without a sign-up. It used to know how to select items and designs based on fit and size. People will have a choice in the things they wish to buy and, in the product's, specific design. effective usage of figma was acquired. learned how to solve problems. a knowledge of design thinking. learned how to efficiently create a prototype.

CHAPTER 7

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